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RESEARCH ARTICLE



Quantitative structure–property relationship analysis of the spectrochemical series by employing electronic descriptors from DFT calculations

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ABSTRACT

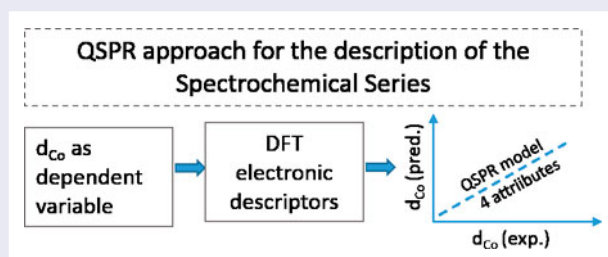
Correlations among Shimura's parameter (d_{Co}) from spectrochemical series for 34 ligands and electronic descriptors from density functional theory calculations are reported. Simple and multiple linear regression models were constructed. The best results were obtained with the descriptors estimated with MN15 functional, obtaining a multiple linear regression model based on a quantitative structure–property relationship (QSPR) approach, with four attributes and the statistical parameters: $R^2 = 0.8543$, $Q^2_{LOO} = 0.8050$, and $Q^2_{LMO} = 0.7977$. The four molecular descriptors are ligand field splitting (Δ), dipole moment (χ_C), enthalpy of the metal complex (H_C) and the ligand HOMO energy (E_{HOMO}). This model obtained for the Ni(0) complexes represents a novel approach to the spectroscopic series description based on computational tools also revising the models proposed from Werner's description until modern recomputations carried out across the spectroscopic series developments. The applicability of the model was evaluated on copper complexes and good prediction of the d_{Co} values for some ligands was also obtained.

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KEYWORDS

Spectroscopic series; ligand field splitting; QSPR; quantum molecular descriptors; metal complexes



Introduction

Spectrochemical Series of the ligands was originally proposed by Shimura [1] based on experimental data, which is in good agreement with the Tsuchida studies [2–4] about the crystal field splitting observed among the e_g (d_{z^2} , $d_{x^2-y^2}$) and t_{2g} (d_{xy} , d_{xz} , d_{yz}) orbitals in transition metal complexes. Shimura predicted the low energy absorptions bands in d^6 -complexes by means of a parameter defined as d_{Co} that was calculated in terms of three component energies of the first-spin d-d absorption allowed band of an octahedral complex. This band was initially explained by Tsuchida finding that the first band was attributed to electron transitions in the unsaturated transition shell of the metal. The shift observed in

the absorption spectra was directly attributed to the ligand effect and its electronic nature i.e. the charge, polarity, and polarisability, when the ligand is coordinated to the metal; the higher electronic parameters, the higher value of P , and therefore the lower of the frequency of the first band is expected. Tsuchida's model explains the three bands observed in UV spectra of transition metal complexes reproducing Equations (1) and (2).

$$S = h\nu_2 = -P - Q + R - \left(\frac{\phi - \sum \phi_i}{n} \right) \quad (1)$$

$$h\nu_3 = S + P + E - J \quad (2)$$

where $S(h\nu_2)$ is the coordination energy (second band), P is the ligand approach energy, Q is the coordination

formation heat, R is the ligand activation energy, ϕ is the lattice energy of the coordination complex, ϕ_i is the lattice energy of the constituent ions and n is the coordination number. The third band energy is $h\nu_3$, where E is the ligand electron affinity and J is the metal ionisation potentials.

Spectrochemical series has been extensively reviewed across literature [5–7] and the one proposed by Shimura [1] is based on experimental d_{Co} evaluations, collecting results from metal complexes with different metals and neutral, charger, bidentate and tridentate ligands.

Density functional theory (DFT) has been applied for the study of the first-row transition complexes [8–10]. Spectrochemical series and the ligand field theory has been addressed by means of ab initio [11–13] and DFT calculations [8,14–23]. Modern aspects in these topics established first by Ballhausen [24] and Jørgensen [6,25] were then deepened by theorists appointing new insights into the molecular electronic structure of the transition metal complex.

Exploring the influence of the parameters in Equations (1) and (2) on the spectroscopic series, we propose an approach from DFT calculations at MN15L and BP86 level, both with Lanl2dz as basis set. Moreover, we include a novel quantitative structure–property relationships (QSPR) model enclosing metal, ligand and complex quantum molecular descriptors (QD) as: ionisation energy (I), electron affinity (A), Mulliken's electronegativity (μ), global hardness (η), global softness (S), global electrophilic power (ω) and thermodynamic data; and we also evaluated the ligand field splitting (Δ), firth excitation energy (EE), Wiberg's Bond Index (WBI) [26] and the recent proposed descriptor Intrinsic Bond Strength Index (IBSI) [27].

Methodology

Software and methods

Homoleptic nickel and copper complexes (d^{10}) within 34 neutral and charged ligands were studied in this work by employing tetrahedral geometry as the initial structures. Gaussian16 [28] and Gassview6.0 [29] software were used to carry out the computational calculations and results visualisations.

Two DFT methods were used to perform the geometry optimisation of complexes: BP86 [30,31] and MN15L [9] functionals, which have been tested as good theory levels to evaluate metal complexes across the literature. The functional BP86 combined with the double-z basis set Lanl2dz has showed good geometry optimisations for first-row transition metals [14] and the theory level selected in this work had been used for the analysis of

closed-shell $Ni(0)L_4$ and $Cu(I)L_4$ homoleptic complexes [32,33]; the basis set 6-31+G(d,p) was also combine with BP86 functional. The recent Minnesota functional MN15L together the basis set Lanl2dz has produced good results for a broad tested database of organometallic compounds reproducing high accuracy electronic and thermochemistry properties for atoms, molecules and solids [9]. The energy difference between occupied and virtual orbitals is a zeroth order approximation to the excitation energy for both functional. On the other hand, the time-dependent density functional theory (TD-DFT) was employed for the analysis of the absorption spectra of the organometallic compounds.

Optimisation and frequency runs

The optimisation was performed by employing the Berny analytical gradient approach with a requested convergence on the density matrix of 10^{-9} atomic units and the convergence criteria for the maximum displacement and maximum force were 0.0018 Å and 0.00045 Hartree/Bohr, respectively. The Hessian matrix, which contains constant forces, was used as the initial information about the nature of the stationary point (minimum or saddle point). All the stationary points were characterised by mean of frequency calculations and from the outputs were extracted the thermodynamic data of complexes, metal and ligands, respectively. From the frequency, outputs were obtained the molecular orbital energy and the isosurface shapes (isovalue = 0.02) for the highest occupied molecular orbital (HOMO), the lowest unoccupied molecular orbital (LUMO), the natural bond orbital (NBO) and the natural localized molecular orbital (NLMO).

Quantum molecular descriptors from FMOs

The descriptors used in this work, directly calculated from HOMO and LUMO frontier molecular orbitals (FMOs), according to the literature are ionisation energy (I), electron affinity (A), Mulliken's electronegativity (μ), hardness (η), softness (S) and electrophilic power (ω) [34]. Polarisability (α) was directly extracted from the frequency output files.

Based on tetrahedral geometry, the ligand field splitting (Δ) for the outputs geometries are calculated as tetrahedral (Td, Equation (3)); square-planar (Sp, Equation (4)) and linear (L, Equation (5)). The MO energy patterns for each geometry are shown in Figure 1.

$$\begin{aligned} \text{For Td: } \Delta &= 219474.63 \frac{\text{cm}^{-1}}{\text{a. u.}} \\ &\times [\text{Avg}(\text{HO} + \text{HO}_{-1} + \text{HO}_{-2}) \\ &- \text{Avg}(\text{HO}_{-3} + \text{HO}_{-4})] \end{aligned} \quad (3)$$

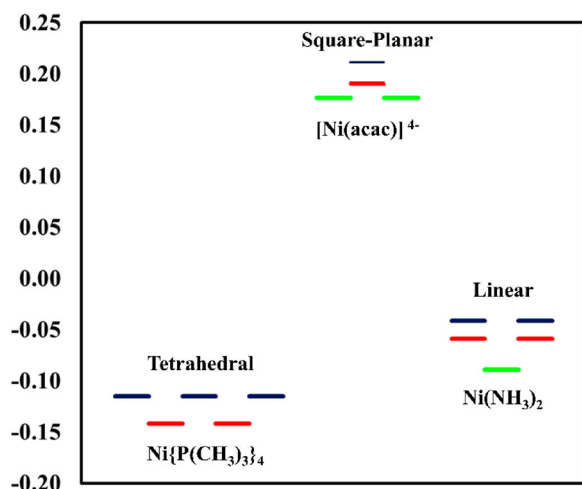


Figure 1. Molecular orbital energy patterns for selected complexes. According to the ligand field splitting calculations from Equations (3)–(5) red and blue lines indicate the splitting of the orbitals.

$$\text{For Sp: } \Delta = 219474.63 \frac{\text{cm}^{-1}}{\text{a. u.}} \cdot \left(\frac{4}{9}\right) \times [\text{Avg}(\text{HO} + \text{HO}_{-1}) - \text{Avg}(\text{HO}_{-2} + \text{HO}_{-3})] \quad (4)$$

$$\text{For L: } \Delta = 219474.63 \frac{\text{cm}^{-1}}{\text{a. u.}} \times \left(\frac{7}{9}\right) \times [\text{Avg}(\text{HO} + \text{HO}_{-1}) - \text{Avg}(\text{HO}_{-2} + \text{HO}_{-3})] \quad (5)$$

In Equations (3)–(5), the HOMOs have been abbreviated as HO for the d-orbitals; the orbital energy average (Avg) has been applied when non-degenerated orbitals are observed.

A TD-DFT calculation was done as a single point from optimised structure. Then, the first vertical excitation energy (EE) was extracted from the output. WBI was obtained doing a single point run and setting a NBO analysis using the NBO3.0 software hosted in Gaussian16.

Intrinsic Bond Strength Index

The IBSI allows to classify chemical bonds by their intrinsic strength starting from the independent gradient model which locally quantifies the electron density from the interpenetration of the wave function calculations [27].

Starting from the optimised structure a single point **wfn** (or **wfx**) file was created in Gaussian. After, an **igm** file was prepared [35] with a text editor as described in Figure 2.

Finally, the **wfn** (or **wfx**) and **igm** files were run together in the IGMPlot2.6.7 [27,35–37] software to generate the IBSI **out** file where the metal–ligand information is stored.

Quantitative structure–property relationship

A QSPR analysis was performed on the dataset based on the possibility of a multivariable description of the spectroscopic series. In this respect, the d_{Co} was used as the response variable and QDs as independent variables. Then, a variables selection was performed by employing the genetic algorithm approach by maximising the value of the cross leave-one out validation coefficient (Q^2_{LOO}) [38]. Only models with a maximum of four non-collinear descriptors were considered, which were well-validated in term of the predictability performance. Two descriptors are considered collinear if the Pearson coefficient in the correlation matrix is ≥ 0.7 [39].

For the model construction, the multiple linear regression (MLR) technique, which is a well-known approach, was widely used for modelling multi variable problems, such as quantitative structure–activity relationship (QSAR) and QSPR [39–42].

The obtained model is validated in terms of some relevant statistical parameters, such as the coefficient of determination (R^2), the cross leave-one out validation coefficient (Q^2_{LOO}), the cross leave-many out-validation coefficient (Q^2_{LMO}) and the bootstrapping coefficient (Q^2_{boot}). Values close to 1 of these coefficients suggest that the model has a good fitting and a good predictability capability. The 5th Setubal Principle to Y-scrambling determination, $a(R^2)$ and $a(Q^2)$, is used to evaluate the possible of predictability by chance of the model, and small values of these coefficients indicate that the model does not predict by chance. The standard deviation error of calculation (SDEC) and the Fisher ratio value (F) are also used to evaluate the fitting of the model.

Results and discussions

DFT geometry optimisation of the nickel homoleptic complexes was performed with each one of the 34 ligands of the spectroscopic series considering neutral, anionic and bidentate ligands. The complexes converged in three different geometries as square-planar (Sp), tetrahedral (Td) and linear (L) (Table 1). The linear geometry was the product of two ligands dissociated, and most of them outputs contain ligands N-bonded or O-bonded to the metal centre (mainly sigma-donor ligands). All frequency runs displayed that the minimum electronic energy was reached. The correlation matrix for the different descriptors obtained for the two functional, BP86

Line 1	1	Default
Line 2	Filename.wfx	Prepared from the frequency output in Gaussian
Line 3	IBSI 3.0	Default = blank; Grid for wfx in Armstrong = 3.0
Line 4	Metal-Atom 1	Bond between the metal and de coordinated atom
Line 5	ENIBSI	Default

Figure 2. Information included in the .igm file used for the IBSI calculation.

Table 1. Convergence and geometry screening for global hardness (η) and ligand field splitting (Δ) at BP86/Lanl2dz and MN15/Lanl2dz theory levels.

Ligand	Geometry	η (eV)		Δ (cm ⁻¹)		d_{Co} (10 ⁻³ cm ⁻¹)
		BP86	MN15L	BP86	MN15L	
CO	Td	4.0426	4.5714	9355.3	8817.8	35.0
^a CN ⁻	Td	0.7252	0.5360	4889.9	4063.9	26.7
P(OCH ₃) ₃	Td	3.4508	4.1048	6310.3	7314.7	29.4
[CH ₂ PMe ₂] ₂	Td	3.2378	3.1532	4624.3	4744.3	26.0
PMe ₃	Td	3.1412	3.0732	5353.5	5946.7	25.6
PH ₃	Td	3.3350	3.3886	4756.4	4766.3	25.6
NO ₂ ⁻	L	1.2168	1.6090	4054.2	3997.9	25.3
NCCH ₃	Td	2.7658	2.8430	5497.8	5208.9	23.5
^b PPh ₃	Td	1.7554	-	3903.4	-	23.2
NH ₂ OH	L	0.7978	1.1110	3047.0	3043.6	22.8
Bpy	Td	0.7216	0.8980	6457.3	5419.2	22.2
Phen	Td	0.7044	0.9048	6489.9	5709.6	22.0
^c Im	Td/L	1.0474	0.9766	4679.7	2940.4	21.2
NH ₃	L	0.6288	0.6436	4312.8	2980.5	21.0
Py	Td	1.0530	1.2236	6037.2	5713.5	20.8
NH ₂ CHO	L	2.6226	2.9534	1597.8	1713.9	20.7
NCNH ₂	Td	1.9426	1.9074	4622.9	3875.2	20.2
NH ₂ CH ₃	L	1.0046	0.9806	4145.5	2730.4	20.0
acac ⁻	Sp	0.6030	0.7760	1774.5	1380.5	16.9
H ₂ O	L	0.8450	0.9192	4135.3	2544.3	16.5
NCS ⁻	Td	1.5148	0.9246	3628.3	3206.5	16.3
CH ₃ OH	L	0.9450	1.0028	4050.8	2439.3	16.2
NCO ⁻	L	0.2136	0.0614	3974.0	1500.5	15.8
OCOMe ⁻	L	0.3648	0.2190	2483.7	462.6	15.5
OH ⁻	Sp	0.3500	-	1420.0	2013.5	14.5
OCHNMe ₂	L	1.0188	1.3666	3149.5	2127.0	15.3
NO ₃ ⁻	L	0.6338	0.8000	2870.4	2356.5	15.2
Ur	L	1.3896	1.3720	2851.6	1672.0	15.15
F ⁻	Td	0.5534	0.4134	3020.0	2024.3	14.8
OSMe ₂	L	0.9138	1.0398	2425.7	2641.6	14.0
OCMe ₂	Td	1.2688	1.1942	2239.7	3159.0	14.0
N ₃ ⁻	Td	0.7994	0.3258	1922.6	1251.7	13.9
Cl ⁻	Td	0.4504	0.4156	1486.7	580.5	12.5
Br ⁻	Td	0.2518	4.5714	1009.6	347.9	11.7

^aWe chose the d_{Co} value reported by Jørgensen (Shimura's $d_{Co} = 32.1 \times 10^3$ cm⁻¹).

^bNo convergence at MN15L level.

^cTetrahedral (Td) at BP86 and Linear (L) at MN15L.

and MN15L, was considered for the modelling but convergence problems with the 6-31 + G(d,p) basis set produced a low number of optimised complexes (Table S1) and higher run times. The good linear correlation between the electronic parameters for the same ligands calculated for both basis set (hardness, d-orbital energies, etc.) in the nickel optimised outputs (Figure S1) showed that the MN15/Lanl2dz level describe with a good performance the spectroscopic series, which it is in a good agreement with the literature [9,14,33]. Pearson's correlation coefficients for each pair of descriptors as well as the

correlation of each descriptor with the Shimuras's d_{Co} [1] (response variable) were obtained. The highest Pearson's coefficients ($r > 0.73$) with respect to the d_{Co} for each functional were obtained for first excitation energy (EE), global hardness (η) and ligand field splitting (Δ). The descriptors EE and η are collinear. Then, in Table 1 are presented the global hardness (η) and ligand field splitting (Δ) for each complex at the two levels of theory, respectively.

The descriptors EE, Δ and η can be associated with the coordination energy and the absorption bands energies

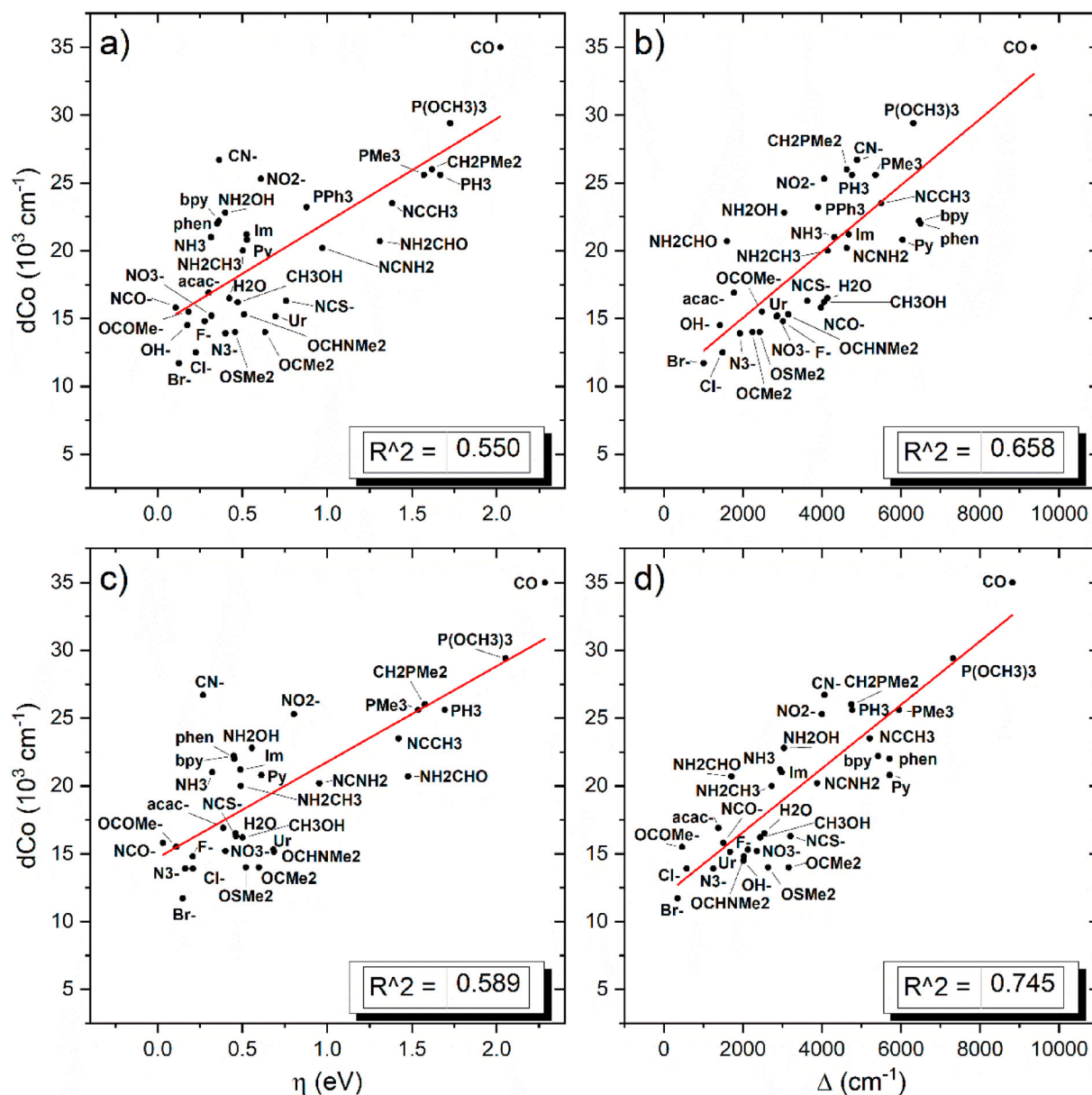


Figure 3. Shimura's d_{Co} as a function of (a) η and (b) Δ at BP86/LanI2dz; (c) η and (d) Δ at MN15/LanI2dz for nickel complexes.

of the transition metal complexes analogously to the work carried out by Tsuchida [4] and considering some aspects of Jørgensen's works [6,25].

The good linear correlations between EE and η are presented for each functional for the nickel cases, consequently, only the graphs containing η and Δ were selected onward considering the experimental response variable as d_{Co} for the simple linear regression models (Figure 3).

Low computational costs for generalized gradient approximations (GGA) density functional BP86 can be obtained but its poorly accuracy to predict values related to MO energies of transition metal complexes has been tested against hybrid and meta-GGA functionals [9], besides exchange–correlation functionals

have shown different performance for first-row transition complex overestimating or underestimating energy levels based on the percentage of HF-exchange energy [20]. MN15L functional shows best performance ($R^2 = 0.745$, Figure 3(d)) whereby the resulted outputs at this level are considered further.

Considering only tetrahedral geometry, neutral ligands and monodentate ligands from Figure 3 (Table 1), better correlations are obtained between d_{Co} and Δ ($R^2 = 0.859$, BP86; $R^2 = 0.851$, MN15L), but statistically a reduced input evaluation matrix implies loss of information on the ligand charge, the coordination ligand mode and the metal complex geometry, therefore a multivariable QSPR analysis is proposed considering all

the attributes of the ligands in the spectrochemical series. Besides thermodynamic data involves in the metal complex formation is considered as an input in our model.

Remarkable aspects of our model are supported by the fact that the second absorption band energy from Equation (1) contains a thermodynamic term (R) assigned to ligand excitation which is affected by the electron affinity of the ligand (L) and the ionisation potential of the metal (J) and when the lattice energy terms are dismissed, the coordination number (n) is not a constraint when gas-phase DFT calculations are performed.

QSPR analysis

After an exhaustive search of the better subset of attributes for the nickel complexes, based on the genetic algorithm approach, a model with four variables was obtained and in Equation (6) are presented the descriptors, as well as the corresponding statistical coefficient.

$$d_{Co}(\text{Ni}) = 12.38 + 3.49\Delta + 0.94\chi_C + 6.77H_C + 8.29E_{HO_L} \quad (6)$$

$$R^2 = 0.8543, Q^2_{LOO} = 0.8050, Q^2_{LMO} = 0.7917, Q^2_{boot} = 0.7897, a(R^2) = 0.070,$$

$$a(Q^2) = -0.310, s = 2.214, F = 41.04.$$

where d_{Co} is Shimura's parameter (10^{-3} cm^{-1}), Δ is the tetrahedral ligand field splitting (10^{-3} cm^{-1}), χ_C is the dipolar moment of the complex (Debye), H_C is the enthalpy of the complex (10^{-3} Hartree) and E_{HO_L} = LUMO of the ligand (a.u.).

The values of the coefficients R^2 and Q^2_{LOO} close to one suggest that this model is robust enough for the prediction d_{Co} prediction. The values of $a(R^2)$ and $a(Q^2)$ demonstrate that there is no correlation by chance, and the slow value of the standard deviation (s) and high value of the Fisher's coefficient (F) suggest a good model fitting. As can be observed in these statistical parameters, better results are found when comparing with the single-variable analysis previously described, suggesting that the representation of the spectroscopic series as a multivariable problem is a more reasonable approach for the understanding of the d_{Co} as response variable. In terms of the standardised coefficients of each descriptor, with values of 1.28, 0.31, 0.39 and 0.22, for the descriptors Δ , χ_C , H_C and E_{HO_L} , respectively, it is possible to mention that the more relevant descriptor is the ligand field splitting, and the others three descriptors have almost the same weight working in a complementary way in obtaining a good fitting and consequently in getting a reliable prediction. It is important to highlight that if

any of these descriptors are removed from Equation (6) the fitting is strongly affected, suggesting no redundancy in the descriptors information. These descriptors do not present collinearity (Pearson's coefficients < 0.6).

The presence of these descriptors in the model is physically interpretable, considering analogies among our QSPR model variables and the Equation (1) variables. Parameters Δ and H_C result of electronic interactions between the ligand (E_{HO_L}) and the metal centre (constant), and mainly due to the ligand approaching the coordination site. Probably, the dipole moment of the coordination complex (χ_C) adjusts the value of d_{Co} parameter considering the geometry distortions and the charge difference between the metal and the ligand.

In Figure 4 is shown the plot of calculated vs. experimental d_{Co} values for the nickel complexes, which demonstrate the good fitting of the model. The model was employed to predict the d_{Co} for NC^- , NF_3 , S^{2-} and H^- ligands and values obtained were 17.5, 19.5, 18.2 and 29.5 ($\times 10^3 \text{ cm}^{-1}$), respectively. Although there are no experimental reported d_{Co} values for these ligands, d_{Co} values for NC^- and NF_3 are in the interval for N-bonded ligands ($15.8\text{--}21.0 \times 10^3 \text{ cm}^{-1}$) [1]; the d_{Co} calculated values for S^{2-} are subject to variations from one complex to another [6], thus the behavior of S^{2-} depends on metal charge varying according to its σ -donor and/or π -donor properties, however, the obtained d_{Co} value for S^{2-} ligand is in the interval for S-bonded ligands as SCN^- and SSO_3^{2-} ($17.0\text{--}19.6 \times 10^3 \text{ cm}^{-1}$) [1]; the calculated d_{Co} for H^- ligand is in accordance with the position assigned by Jørgensen [25] along the spectrochemical series because the ligand has only one lone-pair and forms only σ -bonds acting as a strong field ligand.

The above data evaluated by means of Equation (6) reproduce a spectroscopic series shown in Figure 5 and it has been obtained by DFT at mn15l/lanl2dz including predictions for NC^- , NF_3 , S^{2-} and H^- ligands.

The applicability of the model described by Equation (6) was evaluated by calculating the corresponding descriptors for the complexes formed by the same ligands but considering Cu(I) as metal. In Figure 6 is presented the plot for the experimental vs predicted values of the complex considered in the model construction for the case of the copper complexes. In Table S2, the values of the descriptors calculated for the nickel and copper complexes are shown.

The prediction of the d_{Co} parameters for 7 copper complexes, which were not considered in the model construction, was performed. The results are presented in Table 2, and as can be observed the predicted values are closed to the experimental ones, and a $R^2 = 0.6572$ was obtained for the experimental vs predicted values.

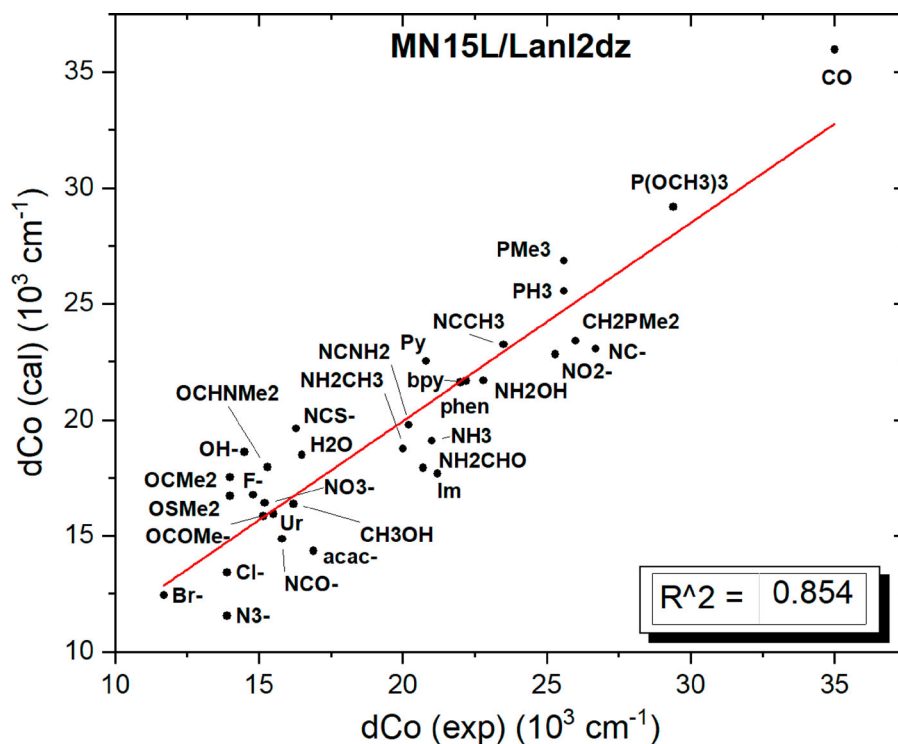


Figure 4. Graphical representation of calculated vs experimental d_{Co} values for the nickel complexes.

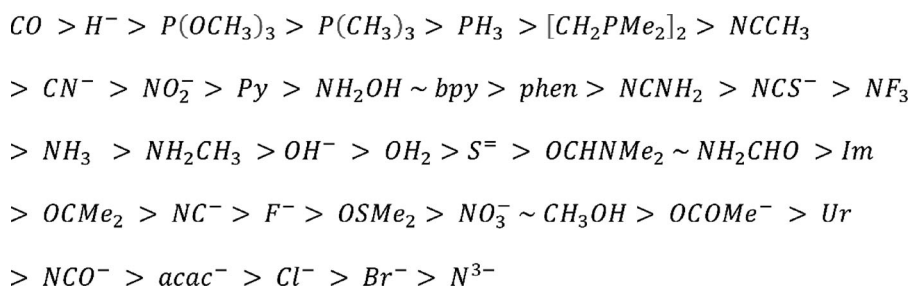


Figure 5. Spectroscopic Series reproduced from DFT-QSPR nickel model at MN15NL/LanI2dz level.

Table 2. Experimental and predicted d_{Co} values for the ligands not considered in the model construction.

Ligand	Geometry	d_{Co} (exp.) 10^{-3} cm^{-1}	d_{Co} (pred.) 10^{-3} cm^{-1}
En	Td	21.4	24.5
ONO-	L	18.2	16.9
Ox	Td	16.6	15.0
O_2^-	Td	15.6	18.5
OCO_2^-	L	15.4	13.0
OSO_3^-	L	15.0	15.6
PPh_3	Td	23.2	21.3

Frontier molecular orbitals and natural population analysis

From the optimised structures for NiL_4 complexes at MN15L level containing $P(OCH_3)_3$, $P(CH_3)_3$, $NCCH_3$ and Cl^- ligands have been selected according to its d_{Co} as a good representation set of the spectroscopic series. Pseudo-tetrahedral geometries have been obtained after

optimisation runs for these complexes. Molecular Orbital energy diagrams show degeneracy of the two orbital sets, t_2 and e_g , for complexes except $NiP(OCH_3)_3$, likely due to tetragonal distortions related to ligand steric effects.

FMOs isosurfaces for the above selected complexes have been plotted (Figure 7) to get insight about the electron delocalisation and the metal–ligand bond covalency. HOMO isosurfaces are strongly delocalised in the bond metal–ligand region for (a)–(c), and the HOMO for (d) is mainly a hybrid orbital ‘d’ of the metal centre; LUMO isosurface for complex (a) is strongly composed by metal–ligand interactions in contrast for (b)–(d), the LUMO is formed mainly by orbitals belonging to ligands. Non-degenerate and high delocalised d-orbitals around the metal centre are fundamental aspects which was taken into consideration to make a correct field splitting

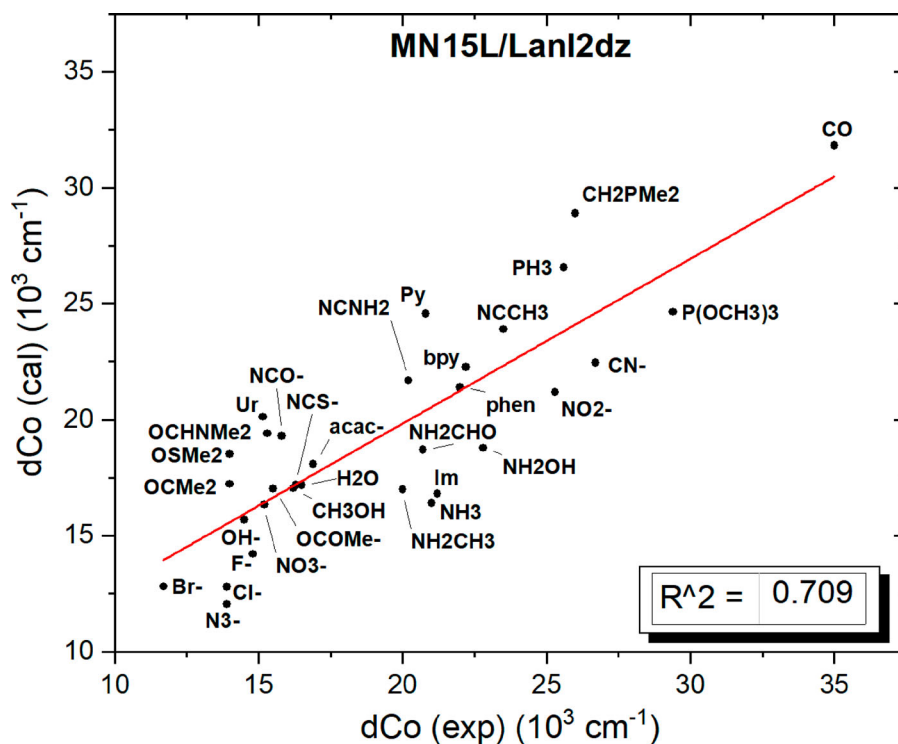


Figure 6. Graphical representation of calculated vs experimental d_{Co} values for the copper complexes.

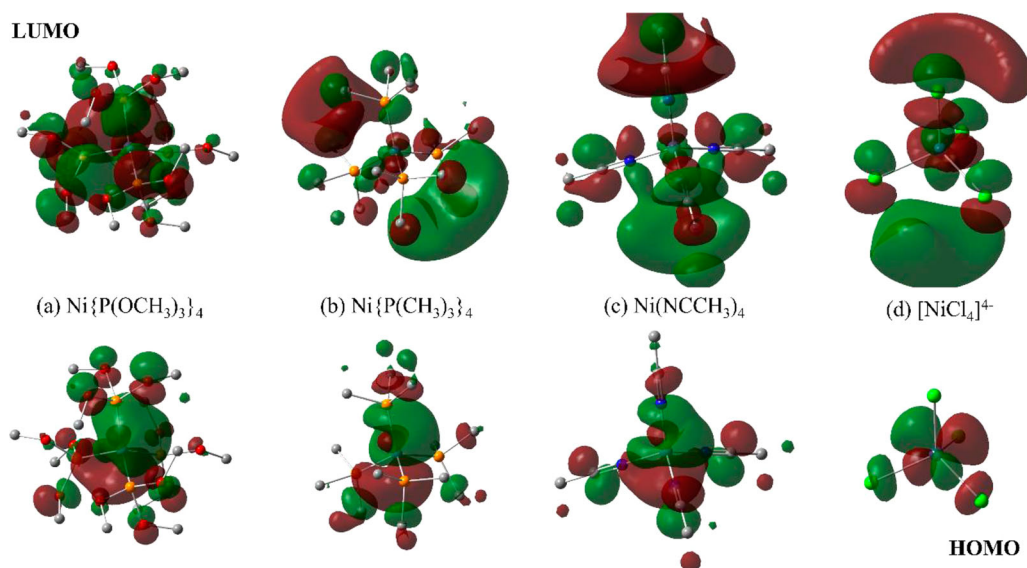


Figure 7. FMOs for tetrahedral NiL_4 complexes: LUMO (top) and HOMO (bottom).

calculation which is the variable that has the greatest weight in the QSPR model (Equation (6)).

Finally, a natural population analysis (Table 3) and NLMO isosurfaces for d-orbitals (Figure 8) were both done to deep insight into other effects involve in the ligand field splitting calculation. From Table 3, both natural electronic configuration and nickel charge leads us to consider that the nickel oxidation state is not strictly zero. Moreover, from the electronic configuration and tetragonal distortions observed, it could be inferred that

Jahn–Teller distortions [43] are present in these complexes which are other factors that affect the value of the ligand field splitting because it should be considered as a correction due to the crystal field stabilisation energy.

NBO-NLMO isosurfaces are shown in Figure 8. In general terms the lower energy NLMOs show less interaction between ligands and the metal centre even when tetragonal distortions are strong, thus d_{z^2} and $d_{x^2-y^2}$ orbitals are easily assigned as unperturbed metal d-orbitals into the classical ligand field splitting calculation

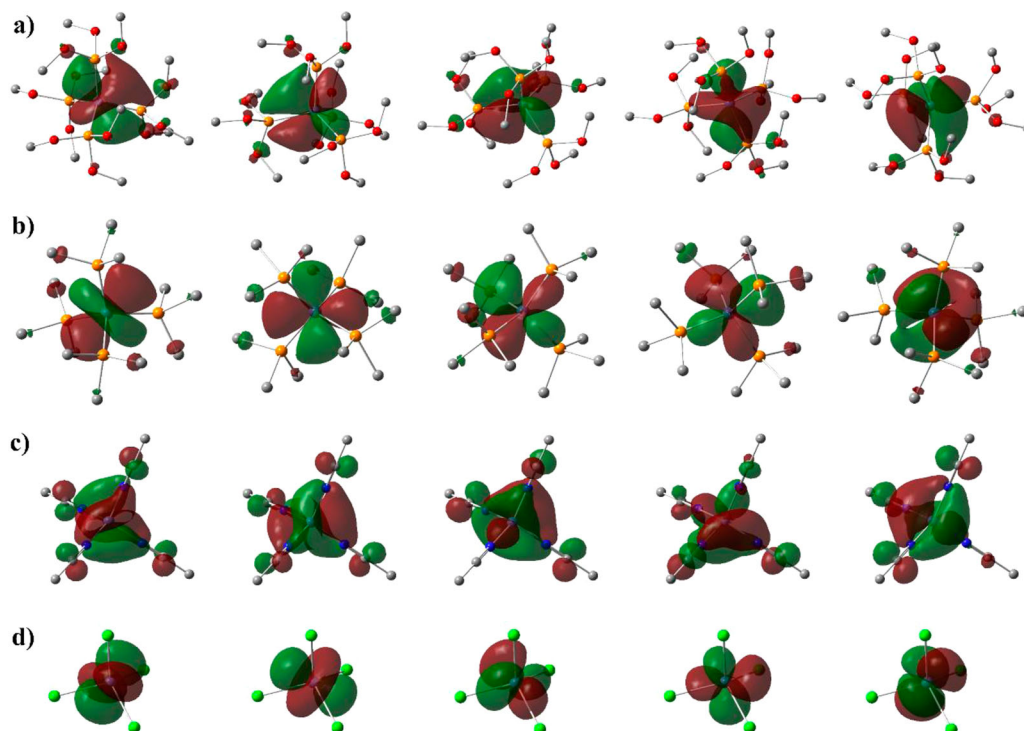


Figure 8. NBO-NLMO isosurfaces at MN15L level for: (a) $\text{NiP}(\text{OCH}_3)_{34}$, (b) $\text{NiP}(\text{CH}_3)_{34}$, (c) $\text{Ni}(\text{NCCH}_3)_4$, (d) $[\text{NiCl}_4]^{4-}$; ordering from low energy (left) to high energy (right) occupied orbitals.

Table 3. Natural population analysis resume for the metal centre.

Complex	Electron configuration [Ni]	Charge [Ni]	Valence	d_{Co}
$\text{NiP}(\text{OCH}_3)_{34}$	$[\text{Ar}]4s^{0.5}3d^{9.4}4p^05p^{1.5}$	-1.357	11.351	29.4
$\text{NiP}(\text{CH}_3)_{34}$	$[\text{Ar}]4s^{0.4}3d^{9.3}4p^05p^{1.4}$	-1.088	11.088	25.6
$\text{Ni}(\text{NCCH}_3)_4$	$[\text{Ar}]4s^{0.2}3d^{9.0}4p^05p^{0.6}$	0.202	9.798	20.0
$[\text{NiCl}_4]^{4-}$	$[\text{Ar}]4s^{0.3}3d^{8.8}4p^{1.8}5p^{0.5}$	-1.358	10.847	12.5

model because these orbitals have components on the axes interacting less with the ligands; on the other hand, higher energy NLMOs show strong ligand-metal interactions due to π -donor (e.g. Cl^- ligand) and π -acceptor (e.g. P-ligands) resulting of metal d-orbital and ligand p-or d-orbitals interactions, which produce a higher hybrid character on d_{xy} , d_{xz} and d_{yz} orbitals and therefore leading to a difficult interpretation of the splitting pattern for Δ calculations. The reader can check some d-orbital splitting patterns reported in the literature for different geometries [18,44–47].

FMOs and NLMOs have been done for $\text{Cu}(\text{I})\text{L}_4$ with the same ligands described in Figures 7 and 8 at MN15L/Lan12z level (Figure S2 and Figure S3). As expected for both metals, an increase in the antibonding interactions between the metal and a strong field ligand (phosphines and acetonitrile) is observed for the HOMO orbitals, and for weak field ligands (halides) the HOMO has a strong d-metal orbital character. In contrast to strong-field ligands, when the metal centre is charged in $\text{Cu}(\text{I})\text{L}_4$ complex with weak field ligands (charged),

the LUMO is not delocalised around copper (Figure S2) and in contrast to the observed in the analogous nickel complex (Figure 7).

Conclusion

The molecular geometry of the optimised molecules demonstrates a high dependency with the nature of the ligand and consequently is difficult to obtain a simple linear regression model based on the field splitting for the description of the spectrochemical series at the DFT level. In this respect, the multivariable approach applied in this article appears to be adequate for the description of the d_{Co} , with good reliability and accuracy. To the best of our acknowledgment, this is the first time that is reported a robust model for the prediction of a big number of ligands (34 in this case) focused on the d_{Co} values. The statistical parameters: $R^2 = 0.8543$, $Q^2_{\text{LOO}} = 0.8050$, $Q^2_{\text{LMO}} = 0.7977$, among others, support our initial hypothesis about the existence of multivariable phenomena involved in the spectrochemical series, involving the four molecular descriptors: ligand field splitting (Δ), dipole moment of the complex (χ_{C}), enthalpy of the metal complex (H_{C}) and the ligand HOMO energy (E_{HOMO}), which are well physically explained in the function of the experimental evidences reported for this systems. The applicability of the model was satisfactorily achieved with the inclusion of the

cooper as the metal centre and also de prediction of other ligands was obtained with a good agreement between experimental and the predicted values.

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